# 4 Functions: representations and reference functions

## Sheet 19 Thu 3 Dec. 2026 *Real-valued function; domain of definition*

**1.** Warm-up · Quick [EQ] Solve $x - 4 = 0$ then $2x + 6 = 0$.

**Ans.** $x = 4$; $x = -3$.

**2.** Core · Method [MOD] Domain of definition of $f(x) = \dfrac{1}{x-4}$?

**Ans.** Exclude the value that makes the denominator zero: $\mathbb{R} \setminus \{4\} = (-\infty, 4) \cup (4, +\infty)$.

**3.** Core + · MCQ [MOD] The domain of definition of $h(x) = \dfrac{1}{x^2+1}$ is:

**A.** $\mathbb{R} \setminus \{1\}$ **B.** $\mathbb{R} \setminus \{-1, 1\}$ **C.** $\mathbb{R}$ **D.** $\mathbb{R}^*$

**Ans.** C. $\mathbb{R}$

*$x^2 + 1 \geqslant 1 > 0$: the denominator is never zero.*

**4.** Review · Table [PNP] Is this table a proportionality table?

| $x$ | 2 | 5 | 8 |
|---|---|---|---|
| $y$ | 6 | 15 | 24 |

**Ans.** Yes: $\dfrac{y}{x} = 3$ for each column.

**5.** Thread · Code reading [ALG] What does $f(4)$ return?

```
def f(x):
    return 3*x - 2
```

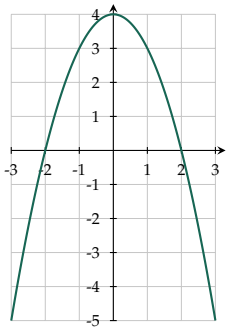
**Ans.** 10.

## Sheet 20 Mon 7 Dec. 2026 *Curve with equation $y = f(x)$; images and preimages*

**1.** Warm-up · Quick [IMG] $f(x) = -x^2 + 4$. Calculate $f(0)$, $f(-2)$ and $f(3)$.

**Ans.** 4; 0; $-5$.

**2.** Core · Graph [IMG] Curve of $f$:




Read off $f(1)$ and the preimages of 0.

**Ans.** $f(1) = 3$; preimages of 0: $-2$ and 2.

*Expression used for the plot: $f(x) = -x^2 + 4$.*

**3.** Core + · Thinking [IMG] For this function, does the equation $f(x) = 5$ have any solutions?

**Ans.** No: $f(x) = 4 - x^2 \leqslant 4 < 5$ for all $x$; the curve does not rise as far as $y = 5$.

**4.** Review · Quick [PRB] Urn: 5 red balls, 3 blue, 2 green. Probability of drawing a blue one? A non-red one?

**Ans.** $\dfrac{3}{10}$; $\dfrac{5}{10} = \dfrac{1}{2}$.

**5.** Thread · Method [EQ] Solve in $\mathbb{R}$: $4 - 3x = 10$.

**Ans.** $x = -2$

*Pitfall: dividing by $-3$ without changing the sign of the result.*

## Sheet 21 Tue 8 Dec. 2026 *Reference functions: absolute value, square, inverse*

1. Warm-up · Quick [CAL] Calculate $|-2.5|$, $(-3)^2$ and $\dfrac{1}{-4}$.

   **Ans.** 2.5; 9; $-0.25$.

2. Core · Table [REG] Complete, for $x = -2\,;\,-1\,;\,0\,;\,1\,;\,2$, the values of $x \mapsto |x|$, $x \mapsto x^2$ and $x \mapsto \dfrac{1}{x}$.

   **Ans.** $|x|$: 2 ; 1 ; 0 ; 1 ; 2; $x^2$: 4 ; 1 ; 0 ; 1 ; 4; $\dfrac{1}{x}$: $-\dfrac{1}{2}$ ; $-1$ ; undefined ; 1 ; $\dfrac{1}{2}$.

3. Core + · MCQ [REG] Which reference curve passes through $(-1,1)$, $(0,0)$, $(1,1)$ and $(2,4)$?

   **A.** the line $y = x$ **B.** the inverse function **C.** the absolute value function **D.** the square function

   **Ans.** D. the square function

   *The absolute value function would pass through $(2,2)$; the inverse function is not defined at 0; the line $y = x$ passes through $(-1,-1)$.*

4. Review · Quick [PRO] An object is enlarged with scale factor 1.5. What does a length of 8 cm become? By what number is the area multiplied?

   **Ans.** 12 cm; the area is multiplied by $1.5^2 = 2.25$.

5. Thread · Prove [LIT] Conjecture: "the sum of two consecutive integers is odd". Prove it.

   **Ans.** $n + (n+1) = 2n + 1$, of the form $2k+1$ with $k = n$: odd.

## Sheet 22 Thu 10 Dec. 2026 *Square root and cube; sign of reference functions*

1. Warm-up · Quick [CAL] Calculate $\sqrt{49}$, $\sqrt{0.25}$ and $(-2)^3$.

   **Ans.** 7; 0.5; $-8$.

2. Core · Method [MOD] Study the sign of $f(x) = -2x + 6$.

   **Ans.** $-2x + 6 = 0 \Leftrightarrow x = 3$; $f(x) > 0$ for $x < 3$, $f(x) < 0$ for $x > 3$ (negative gradient).

3. Core + · True/false [MOD] True or false? Justify. For every real number $x$, $\sqrt{x^2} = x$.

   **Ans.** False. Counter-example: $x = -3$: $\sqrt{9} = 3 \neq -3$. In fact $\sqrt{x^2} = |x|$.

4. Review · Quick [PER] How many squares make up a net of a cube? Does a cross (a row of 4 squares + 1 square on each side) work?

   **Ans.** 6 squares; yes, the cross folds into a cube.

5. Thread · Quick [SOL] How many faces, vertices and edges does a square-based pyramid have?

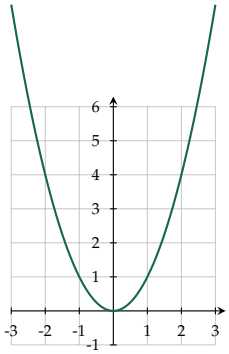
   **Ans.** 5 faces, 5 vertices, 8 edges.

## Sheet 23 Mon 14 Dec. 2026 *Graphical and numerical solving: $f(x) = k$, $f(x) < k$*

1. Warm-up · Quick [EQ] Solve $x^2 = 16$ then $x^2 = -4$.

   **Ans.** $x = \pm 4$; no real solution ($x^2 \geqslant 0$).

2. Core · Graph [REG] Curve of $f(x) = x^2$:




Solve graphically $f(x) = 4$ then $f(x) < 4$.

**Ans.** $f(x) = 4$: $x = -2$ or $x = 2$; $f(x) < 4$: $x \in (-2, 2)$.

*Look for the abscissas of the points of the curve lying below the line $y = 4$.*

3. Core + · Method [EQ] Solve $|x| < 3$ then $|x| = 5$.

   **Ans.** $(-3, 3)$; $x = -5$ or $x = 5$.

4. Review · Pitfall [SUB] Calculate $-x^2$ then $(-x)^2$ for $x = -3$.

   **Ans.** $-(-3)^2 = -9$ and $(-(-3))^2 = 9$.

5. Thread · Method [DEM] In a triangle $ABC$, $M$ and $N$ are the midpoints of $[AB]$ and $[AC]$. What is being proved? Which property is cited?

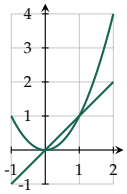
   **Ans.** $(MN) \parallel (BC)$ and $MN = \frac{1}{2}BC$: the midpoint theorem.

## Sheet 24 Tue 15 Dec. 2026 *Solving $f(x) = g(x)$ and $f(x) < g(x)$*

1. Warm-up · Quick [EQ] Solve $2x - 1 = x + 3$.

   **Ans.** $x = 4$.

2. Core · Graph [REG] Curves of $f(x) = x$ and $g(x) = x^2$:




   Solve $f(x) = g(x)$ then $f(x) < g(x)$.

   **Ans.** $x = x^2 \Leftrightarrow x(x-1) = 0$: $x = 0$ or $x = 1$; $f(x) < g(x)$: the parabola is above the line for $x < 0$ or $x > 1$.

   *Intervals:* $(-\infty, 0) \cup (1, +\infty)$.

3. Core + · Code reading [ALG] What does `f(-1) == g(-1)` return? And `f(0) == g(0)`?

```
def f(x):
    return x*x
def g(x):
    return 2*x + 3
```

   **Ans.** `True` ($1 = -2 + 3$) then `False` ($0 \neq 3$): $-1$ is a solution of $f(x) = g(x)$, 0 is not.

4. Review · Problem [MOD] Plan A: 15,000 Ar fixed + 1,000 Ar per session. Plan B: 2,500 Ar per session. From how many sessions is A cheaper?

   **Ans.** $15{,}000 + 1{,}000x < 2{,}500x \Leftrightarrow x > 10$: from 11 sessions (10: equal at 25,000 Ar).

5. Thread · Quick [REP] Calculate the coordinates of the midpoint of $[AB]$ with $A(-2, 5)$ and $B(6, 1)$.

   **Ans.** $\left(\frac{-2+6}{2}; \frac{5+1}{2}\right) = (2, 3)$.